**You provide**

- **reads per animal** — any platform
- **AID-null controls** — two or more

+ a sample sheet

**SHMtrace** — calls and tests against the controls

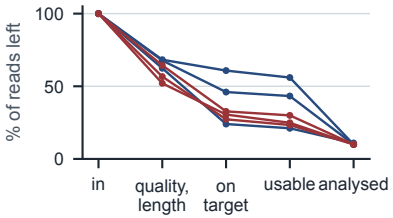
**You get**

- **report.html** — every result on one page
- **figures** — PNG, SVG or PDF
- **tables** — Excel and CSV, every number
- **{ } project.json** — reopens in the app

Examples redrawn from one run (JH4 intron, nanopore, 500 reads per mouse): wild type · AID-null (*Aicda−/−*)

## A Did enough reads get through?

*qc_waterfall*



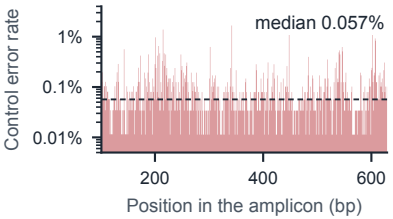

One line per library: the share of reads left after each filter. The last step is the depth you set, not a loss.

**Watch for:** a library far below the others.

**Also:** coverage · quality_histogram · saturation

## B Are the calls real?

*control_background*



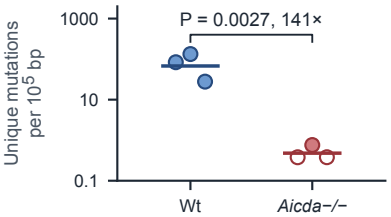

The error rate the AID-null controls showed at each position. A mutation is called only if it clears this floor, with predicted false calls held under 5%.

**Watch for:** spikes: calls there need much more support.

**Also:** volcano · single_read_excess · replicate_concordance

## C How much mutation?

*density_by_group*



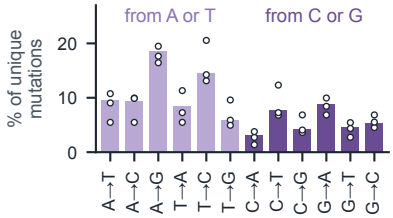

One dot per animal (open: none found); bars are geometric means. The test counts each animal once, however many reads it gave.

**Watch for:** frequency counts a clone's mutations in every read; unique density counts them once.

**Also:** mutation_frequency_by_group · mutated_reads_by_group · mutation_load · combined_density · indels

## D What kind of mutation?

*spectrum_by_group*



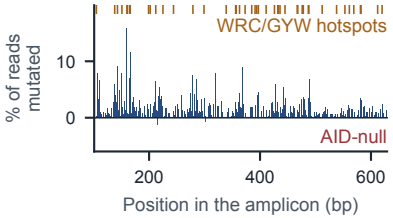

The 12 base changes: bars for wild type, dots per mouse. AID leaves C→T and G→A at WRC/GYW motifs (54% of G:C mutations here).

**Watch for:** compare amplicons on the composition-corrected spectrum.

**Also:** substitution_matrix · hotspot_targeting · spectrum_corrected · targeting_spectrum · mutability_hedgehog

## E Where along the amplicon?

*positional_map*



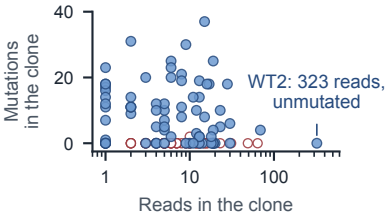

The share of reads mutated at each position: wild type up, AID-null down. Peaks on hotspot ticks are the signature of AID.

**Watch for:** a peak in the controls too: a germline variant or an artefact.

**Also:** positional_by_sample · targeting_profile

## F Which clones carry them?

*clone_map*




Clones from the V(D)J junction; each dot is one clone. In WT2, one unmutated clone held 81% of reads.

**Watch for:** one clone repeating its mutations in many reads.

**Also:** clonality · phasing_blocks

Other routes: a single sample from the command line writes spectrum, spatial and mutation_load; the legacy compare writes three more.